

it up with their respective missions. China is the latest entrant with its Chang-E orbiting the moon's surface since 2007 along with Japan's third moon orbiter, Selene.

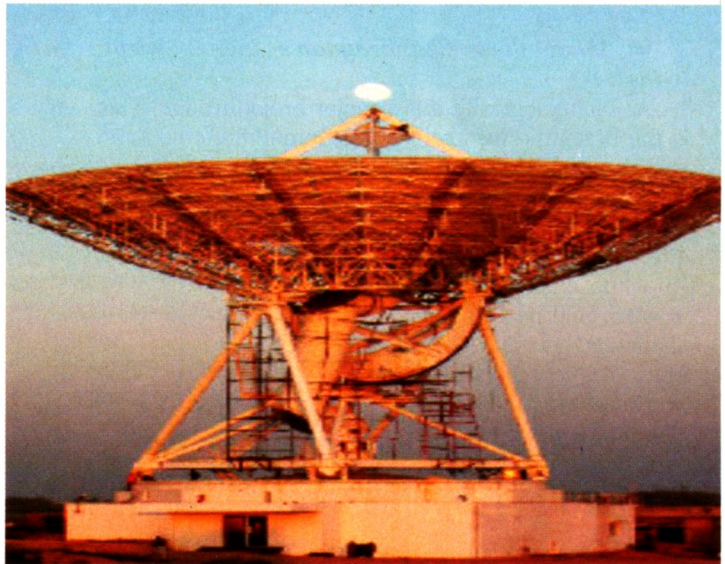
Chandrayaan will not be lonely in its lunar orbit. NASA is keen that it succeeds because it wants to set up a space station on the moon and use it for launching its future manned missions to Mars. It's probing the surface for water ice so that it can tap the hydrogen and oxygen and use it as an unlimited supply of fuel for its future rockets. NASA will send its own lunar orbiter next year to map the terrain more accurately. Spudis says, "The moon is the first stepping stone from the earth to the solar system. If you are going to live in a place, you need a real good map to figure out which is the best area." With enormous funds needed for planetary explorations, international cooperation is imperative both to share costs and pool expertise. Nair points out, "From that angle, Chandrayaan is a good beginning because we are setting an example of how future space explorations could be conducted."

Having kicked off its first lunar mission, ISRO isn't going to be satisfied with a one-off orbiter. It has already got sanction for building Chandrayaan-2 which will have a lander as well as a rover to analyse samples and is targeted to go up somewhere around 2012. It's also planning a Mars orbiter in the near future. Meanwhile, the Space Commission has ap-

total budget. World space powers do acknowledge that ISRO is one of the few organisations that focuses on utilising space resources for development. But Nair says it is increasingly difficult to attract talent and the lunar mission is one way of getting the best to join ISRO. Also with deep space exploration undergoing a major revival, ISRO doesn't want to be left behind.

After having mastered the technology for building giant rocket launchers and state-of-the-art communication and remote sensing satellites, ISRO needs a new challenge to keep its scientists at the frontiers of space. The spin-off

COURTESY: ISRO



THE MOON is the first stepping stone from the earth to the solar system. Chandrayaan will help find the best place for human colonies.

EARS ON THE GROUND: The large antenna in Byalalu that tracks Chandrayaan

proved of a plan for ISRO to begin work on manned space flights. The estimated cost would be Rs 12,000 crore for the mission and it is expected to be launched by 2015. (see box)

Despite the excitement over Chandrayaan's lift-off, certain sections of India's scientific community are unhappy with the mission. They charge ISRO with trying to reinvent the wheel and spending big on planetary explorations and manned missions. Nair insists that ISRO has not deviated from its objectives of providing utilitarian services like communication, weather predictions and disaster warning systems. And that space exploration is less than 5 per cent of ISRO's

benefits from space explorations are enormous and have a major effect on the scientific development of the country. ISRO has demonstrated that a Government organisation tasked with a mission can deliver with extraordinary success. As Nair says, "We are regarded as a developed country where space is concerned."

Of course, its reputation would be on the line in the next fortnight as it makes the intricate manoeuvres to get Chandrayaan to the moon. Herman Andersson of the Swedish Institute of Space Physics, which has an instrument on board the spacecraft, summed up the situation when he said, "So far, so good." ■